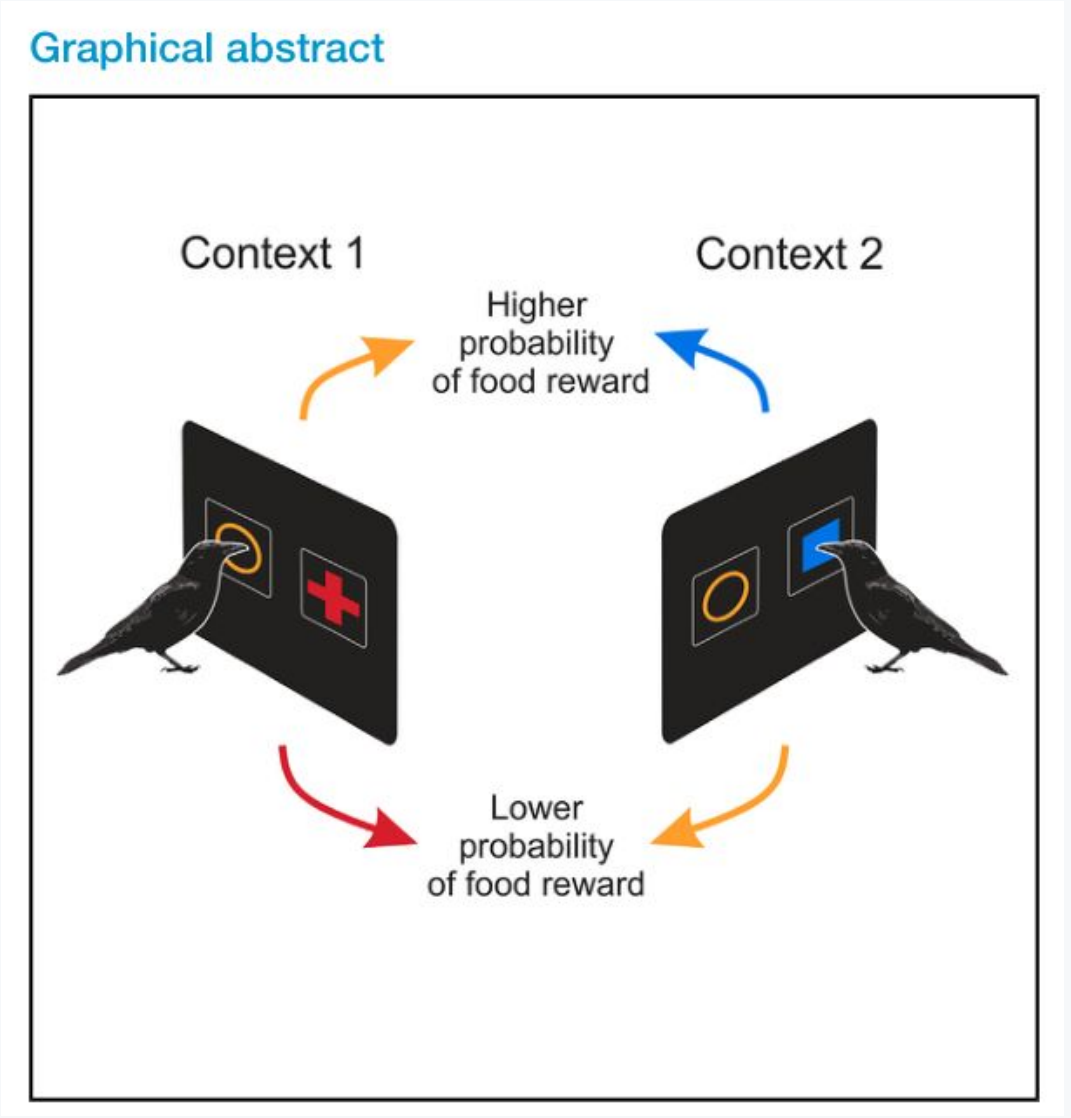
Do crows think in probabilities?

A general-audience presentation of Johnston, Brecht, & Nieder (2023)

Main idea: carrion crows learned which symbols were more likely to lead to food, then used that memory to make better choices in new situations.



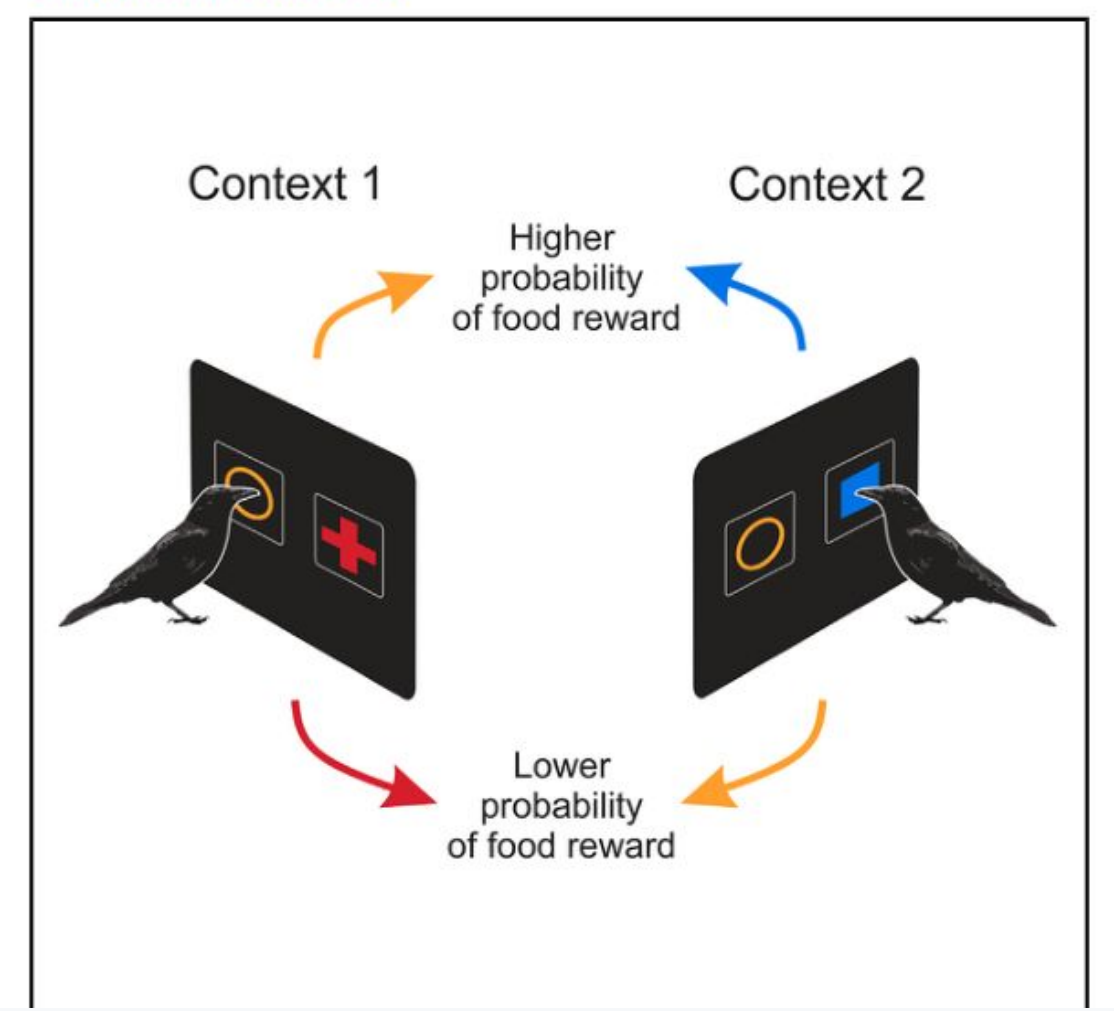
Study: “Crows flexibly apply statistical inferences based on previous experience”

Background: what is statistical inference?

Statistical inference means using limited past evidence to estimate what is likely to happen next.

- Everyday example: if one snack jar gives your favorite candy more often than another jar, you will usually pick that jar next time.
- The key idea is probability: the chance that an outcome will happen.
- A harder version of this task is remembering those chances from earlier experiences rather than seeing the evidence right in front of you.

Why it matters
Animals often forage under uncertainty, so estimating which option is most rewarding can improve survival.



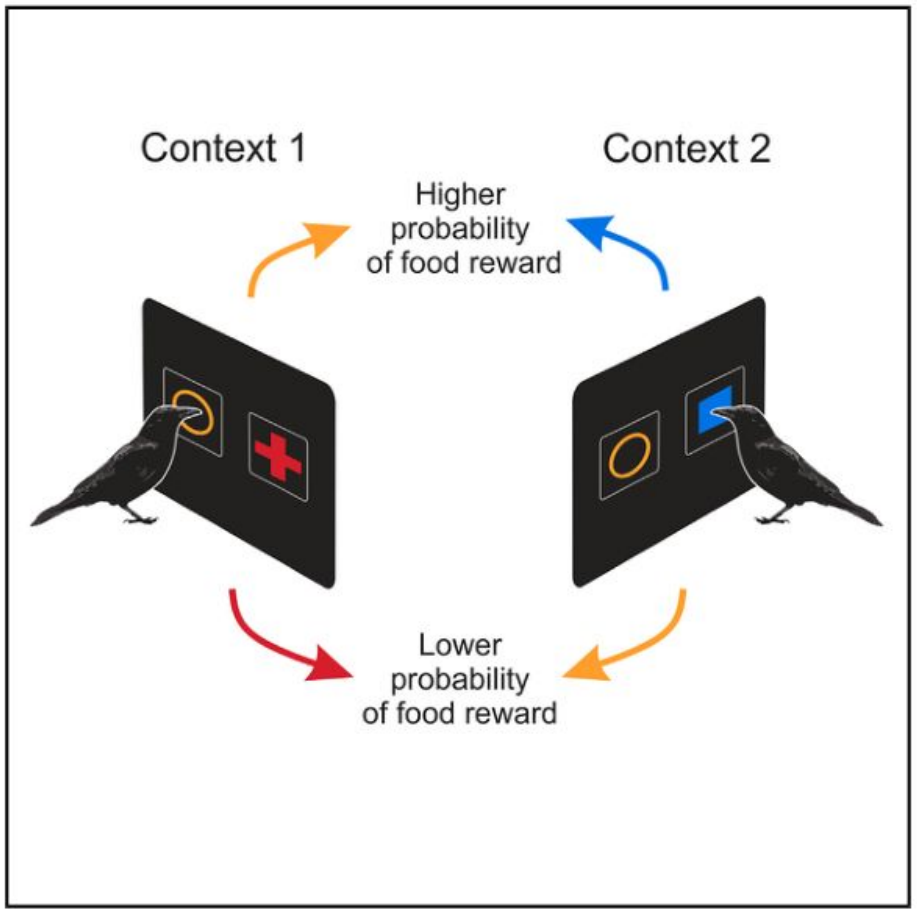
Why test crows?

The study used two male carrion crows (Corvus corone), ages 7 and 8.

- Carrion crows are corvids, a bird group known for flexible problem-solving and strong memory.
- These birds had previous experience with touchscreen tasks, so they could learn symbol–reward links efficiently.
- Researchers chose crows because they already show abstract visual learning and planning abilities.

Research question
Can crows remember several reward probabilities and flexibly choose the better option in new pairings?

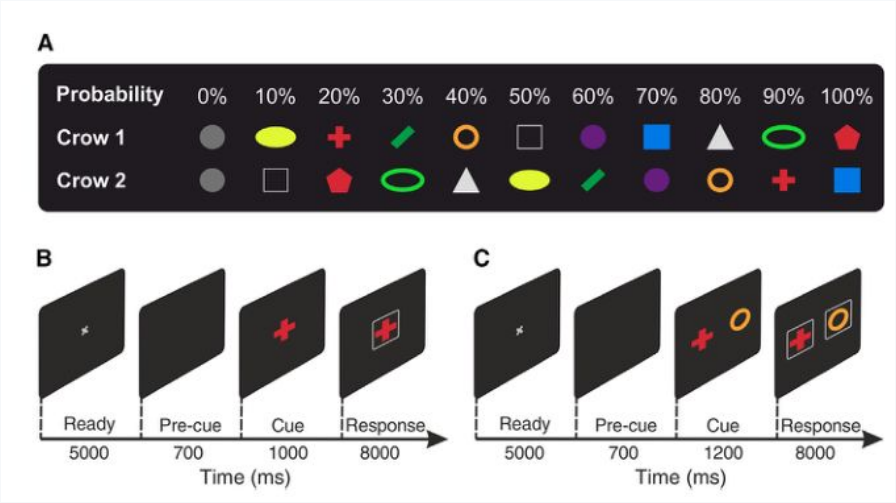
Graphical abstract



Methods: how the experiment worked

Step 1: learn the symbols

- Each crow learned 9 different symbols.
- Every symbol had its own chance of giving food: 10%, 20%, 30% ... up to 90%.
- The symbols themselves were arbitrary, so the birds had to learn from experience rather than from any obvious visual clue.



Step 2: forced-choice test

- Two symbols appeared together on the screen.
- The crow pecked one symbol.
- A choice counted as optimal when the crow picked the symbol with the higher reward probability.

Design detail to know: a “forced-choice” task simply means the animal must choose between the options shown.

Important point: the birds had to rely on memory from previous trials, not on visible proportions at test time.

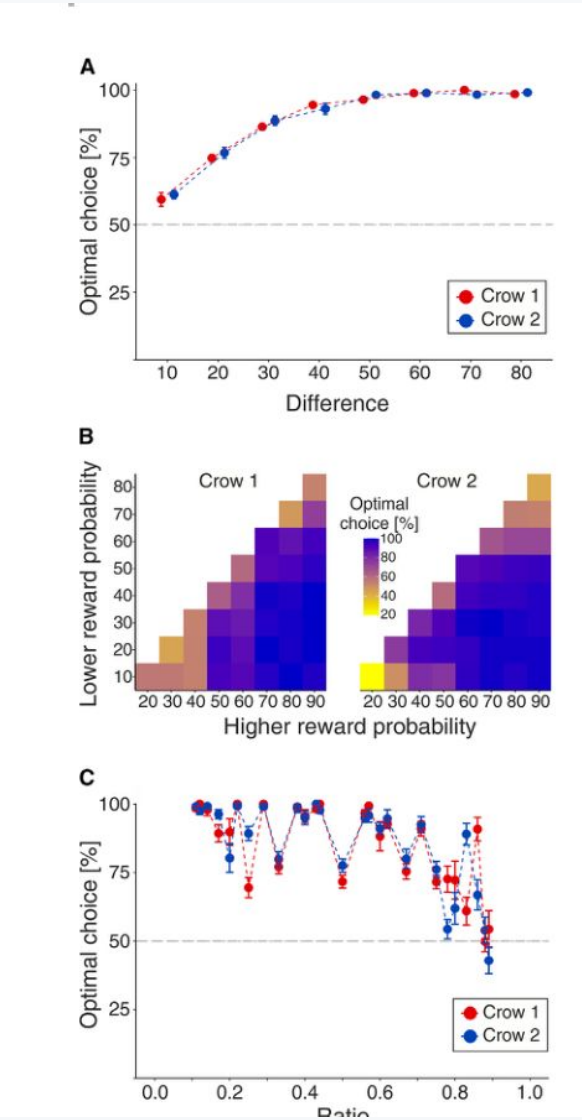
Results: the crows usually picked the better option

Main finding

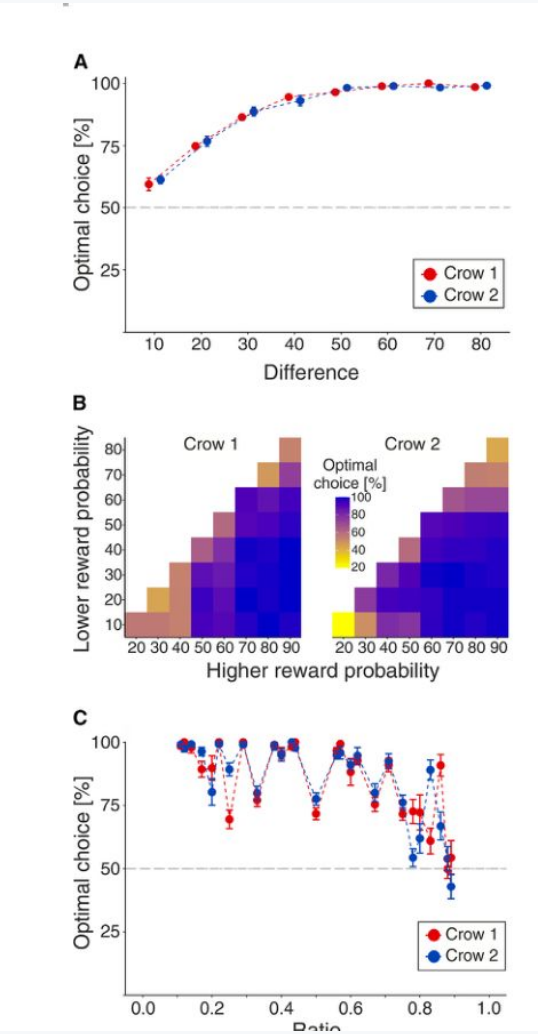
Both crows chose the higher-probability symbol well above chance from the start of testing.

- Chance performance would be 50% if the birds were guessing.
- Instead, each crow was at or above 76% optimal choices across sessions.
- They also stayed accurate when retested about a month later, suggesting the probabilities were remembered over time.

In plain language: the birds learned which symbols were more reliable “bets.”



Results: the birds used probabilities flexibly



Two patterns mattered:

- Distance effect: decisions were easier when the difference between two probabilities was larger (for example, 90% vs 10% is easier than 60% vs 50%).
- Magnitude/ratio effect: even the same absolute gap was not always equally easy. The ratio between the two values also mattered.
- These are classic signatures of approximate magnitude thinking: the birds seemed to treat probability as a mental quantity they could compare.

Why this matters
The crows were not just memorizing “good” and “bad.” They compared probabilities in context.

Control test: relative chance vs. raw reward count

Researchers added a control to rule out a simpler explanation.

- One new symbol paid off 80% of the time.
- Another paid off 40% of the time but was shown twice as often during training.
- That meant both symbols produced a similar total number of rewards overall.

Result: the crows still preferred the 80% symbol.
So they were using relative probability, not just counting which symbol had delivered more food in total.

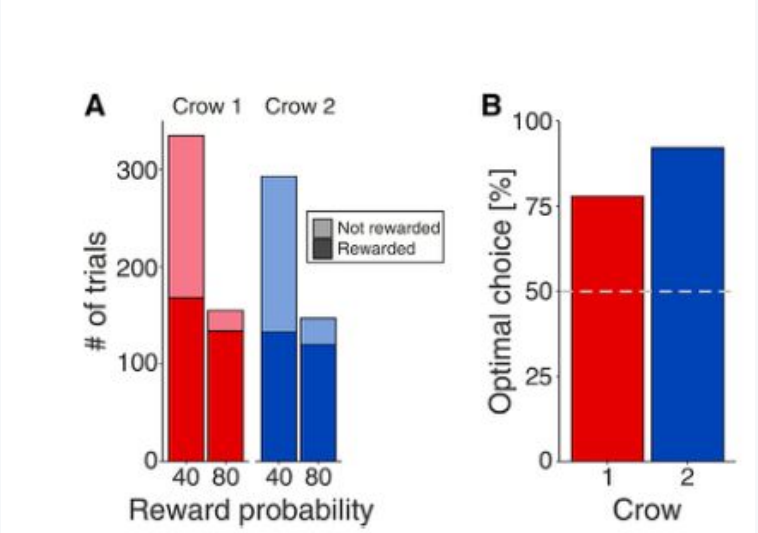


Figure 4. Testing absolute versus relative reward frequency i control task

(A) The absolute frequency crows were rewarded during training with tw stimuli. One stimulus was associated with an 80% reward probability a other with a 40% reward probability, with the latter being presented tw often.

(B) Percentage of trials in which the crows made the "optimal" choice sponding to the stimulus associated with the higher reward probability the forced-choice paradigm with the newly trained stimuli. Dashed lin resents chance performance.

Discussion: what does this mean in the real world?

What the study suggests

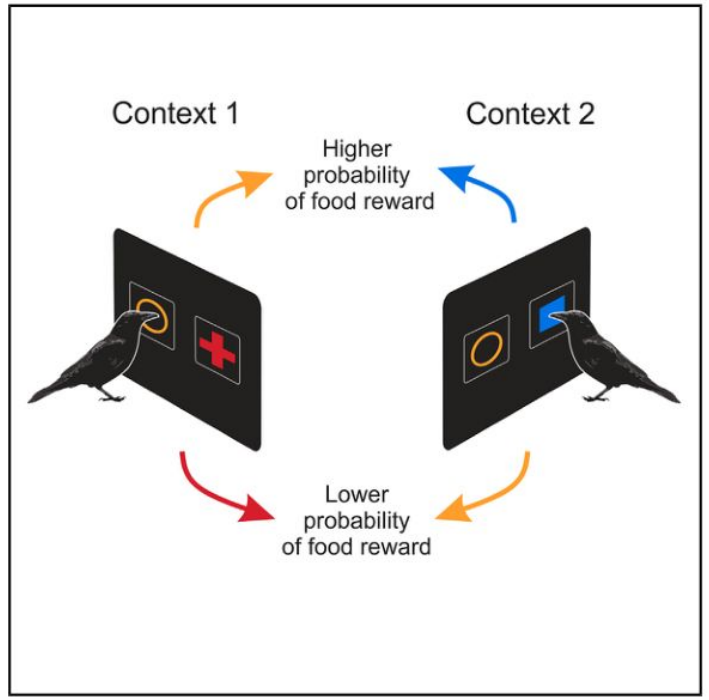
- Crows can store several learned probabilities in memory.
- They can compare those remembered probabilities in new pairings.
- Their behavior looks like a simple form of statistical reasoning under uncertainty.

Why that matters beyond crows

- It adds to evidence that sophisticated reasoning is not uniquely human.
- It helps explain how animals may make adaptive foraging decisions when outcomes are uncertain.
- It also links probability judgments to a broader “analog magnitude” system used for comparing quantities.

Bottom line
The birds did more than memorize a single rule; they used remembered probabilities in a flexible, context-dependent way.

Graphical abstract



Critical evaluation: what the study did well

1. Strong experimental control

- Arbitrary symbols reduced the chance that birds could rely on obvious visual cues.
- The control task directly tested a simpler alternative explanation: total reward count.

2. Flexible reasoning was tested, not just simple memory

- Using 9 probabilities and many pairings made the task more demanding than a basic two-choice setup.
- Finding distance and ratio effects strengthens the argument that crows represented probability as an abstract magnitude.

Why these are effective design choices

- They improve internal validity: the study more clearly measures what it claims to measure.
- They also make the conclusion more persuasive for a skeptical reader: the researchers tested competing explanations instead of ignoring them.

Course concept connection: this is a good example of careful operationalization and control of confounds.

Critical evaluation: limitations and open questions

1. Very small sample

- Only two crows were tested.
- That is common in animal cognition research but it limits generalizability.

2. The birds were unusual compared with wild crows

- They were hand-raised, lab-housed, and already trained on touchscreen tasks.
- So the results may not perfectly reflect what naïve or wild crows would do.

3. What the study cannot fully tell us

- It shows behavior consistent with statistical inference, but it cannot reveal the birds' exact mental strategy with certainty.
- Future studies could test more birds, other species, or more naturalistic foraging tasks.

A balanced conclusion

- The paper makes a strong case for flexible probability-based choice in these two trained crows.
- But the evidence is still best interpreted as promising and impressive rather than the final word for all birds.

Reference

Johnston, M., Brecht, K.F., & Nieder, A. (2023). Crows flexibly apply statistical inferences based on previous experience. Current Biology, 33, 3238–3243.